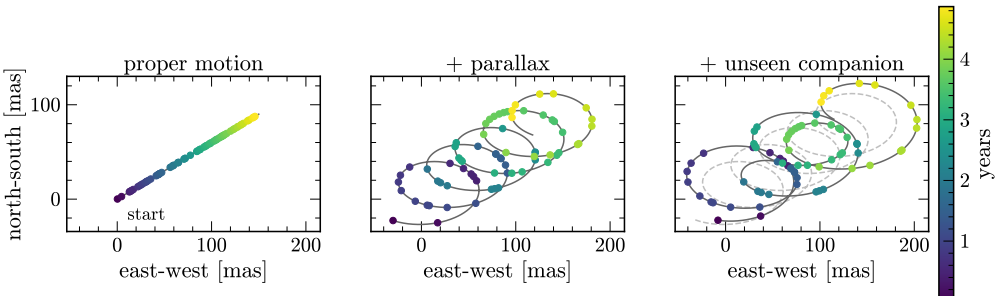
Planets or binary stars?

Finding planets with Gaia, and the impostors that look like them

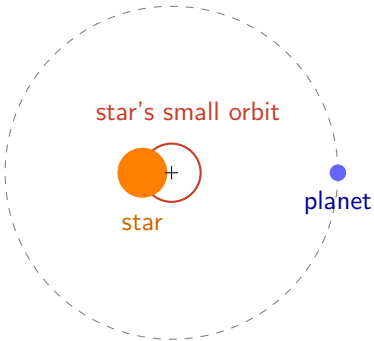
SRMP 2026

Recap: the path of a star



Straight-line drift, plus a yearly loop from parallax. An unseen companion adds a wobble. That wobble is how Gaia finds planets.

Planets make their star wobble

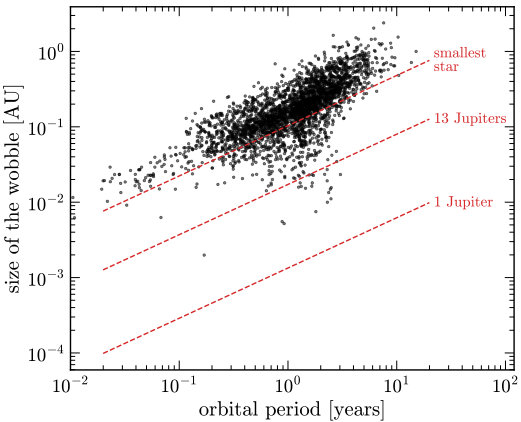


- The planet doesn't just orbit the star. Both orbit their shared balance point (center of mass).
- So the star moves in a small circle too.
- Bigger wobble for: heavier planets, wider orbits, lighter stars, closer stars.

seen from 10 pc	wobble
Sun, pulled by Jupiter	500 μas
Sun, pulled by Earth	0.3 μas
Gaia's best	$\sim$20 μas

1 μas = 0.001 mas. Gaia can see Jupiters, not Earths.

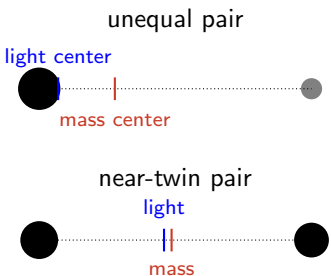
Wobbles found in DR3



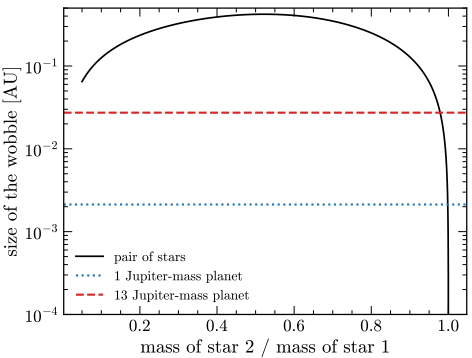
- Each dot: a star within 100 pc where Gaia found an orbit. Wobble size in AU (1 AU = Earth–Sun distance).
- Dashed lines: the wobble a dark companion would cause, for a star a bit smaller than the Sun. A planet would sit below the 13-Jupiter line.
- Almost all of these are pairs of stars, not planets.
- DR3 only published the clearest orbits. DR4 will release every measurement, so we can search all stars ourselves.

The problem: impostors

- Two stars that are too close together look like one dot to Gaia.
- Gaia sees the center of their *light*. Both stars orbit the center of their *mass*.
- Heavier stars are much brighter, so the light center sits close to the heavier star. Unequal pair: a big wobble, an obvious binary.
- If the two stars are almost twins, the two centers almost coincide. The wobble is tiny, and looks just like a planet.



How common are impostors?



- To fake a planet, the two stars must have almost the same mass: within 2% to fake a heavy planet, within 0.2% to fake Jupiter.
- That sounds rare, but pairs of stars are far more common than giant planets.
- When astronomers checked 20 DR3 wobbles that looked like planets or brown dwarfs ("failed stars"), 13 turned out to be twin stars.
- DR4 is expected to find thousands of planets. Some of them will be impostors.

Our project

How can we tell a real planet from a pair of twin stars,
using only Gaia data?

Some ideas:

- Twin stars give twice the light of one star of the same colour. So they should sit above the main sequence on the HR diagram. Can we use that?
- Does the shape or size of the wobble give it away?
- Can we simulate both kinds of systems and learn to tell them apart?

Plan: learn the tools, build simulations, and be ready when DR4 comes out.

References

- Perryman et al. 2014, *Astrometric exoplanet detection with Gaia*, ApJ 797, 14
- Gaia Collaboration, Arenou et al. 2023, *Gaia DR3: Stellar multiplicity, a teaser for the hidden treasure*, A&A 674, A34
- Holl et al. 2023, *Gaia DR3 astrometric orbit determination with MCMC and genetic algorithms*, A&A 674, A10
- El-Badry et al. 2024, *A generative model for Gaia astrometric orbit catalogs*, OJAp 7, 100
- Stefánsson et al. 2025, *Gaia-4b and 5b*, AJ 169, 107
- Marcussen et al. 2026, *Radial velocity follow-up of Gaia astrometric substellar companions: six confirmed brown dwarfs and 13 impostor binaries*, A&A, in press
- Lammers & Winn 2026, *On the exoplanet yield of Gaia astrometry*, AJ 171, 18

Orbit plots made from Gaia DR3 data with the tutorial notebooks.